

Лабораторная работа №5. Менеджер паролей pass и управление файлами конфигурации

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1. Цель работы

- Изучить работу менеджера паролей pass в Unix-системах
- Освоить использование GPG для шифрования данных
- Понять принципы организации структуры паролей
- Настроить синхронизацию через Git
- Изучить инструмент управления конфигурациями chezmoi
- Научиться применять шаблоны для адаптации конфигураций

2. Ход работы и результаты

2.1 Подготовка окружения

Установка инструментов

```
newy
Fetched 3500 kB in 2s (1452 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
118 packages can be upgraded. Run 'apt list --upgradable' to see them.
(Chase) libqrencode-bin@ubuntu:/work/study/2026-2027/Onepassword/cacremu/os-intro/labs/lab05$ sudo apt install pass -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  dmcc libdecor-0 libdecor-0-plugin-1-gtk libgsoap-2.8.132t64 liblsbf libqt5help5 libqt5opengl5t64 libqt5printsupport5t64 libqt5sql5-sqlite libqt5sql5t64 libqt5x11extras5
  libsamplerate0 libstdc++2.0-0 libtss0 libvncserver1 linux-headers-6.8.0-101 linux-headers-6.8.0-101-generic linux-headers-generic
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  libqrencode4 qrencode wl-clipboard xclip
Suggested packages:
  libxsl-simple-perl python
The following NEW packages will be installed:
  libqrencode4 pass qrencode wl-clipboard xclip
0 upgraded, 5 newly installed, 0 to remove and 118 not upgraded.
Need to get 137 kB of archives.
After this operation, 478 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/universe amd64 libqrencode4 amd64 4.1.1-1build2 [25.0 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble/universe amd64 pass all 1.7.4-6 [34.5 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble/universe amd64 qrencode amd64 4.1.1-1build2 [26.1 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble/universe amd64 wl-clipboard amd64 2.2.1-1build1 [33.4 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble/universe amd64 xclip amd64 0.13-3 [17.6 kB]
Fetched 137 kB in 0s (593 kB/s)
Selecting previously unselected package libqrencode4.
(Reading database ... 380382 files and directories currently installed.)
Preparing to unpack .../libqrencode4_4.1.1-1build2_amd64.deb ...
Unpacking libqrencode4 (4.1.1-1build2) ...
Selecting previously unselected package pass.
Preparing to unpack .../archives/pass_1.7.4-6_all.deb ...
Unpacking pass (1.7.4-6) ...
Selecting previously unselected package qrencode.
Preparing to unpack .../qrencode_4.1.1-1build2_amd64.deb ...
Unpacking qrencode (4.1.1-1build2) ...
Selecting previously unselected package wl-clipboard.
Preparing to unpack .../wl-clipboard_2.2.1-1build1_amd64.deb ...
Unpacking wl-clipboard (2.2.1-1build1) ...
Selecting previously unselected package xclip.
Preparing to unpack .../xclip_0.13-3_amd64.deb ...
Unpacking xclip (0.13-3) ...
Setting up libqrencode4 (4.1.1-1build2) ...
Setting up qrencode (4.1.1-1build2) ...
Setting up xclip (0.13-3) ...
Setting up pass (1.7.4-6) ...
Processing triggers for man-db (2.12.0-0build2) ...
Processing triggers for libc-bin (2.39-0ubuntu7) ...
(Chase) libqrencode-bin@ubuntu:/work/study/2026-2027/Onepassword/cacremu/os-intro/labs/lab05$ sudo apt install pass-otp -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  dmcc libdecor-0 libdecor-0-plugin-1-gtk libgsoap-2.8.132t64 liblsbf libqt5help5 libqt5opengl5t64 libqt5printsupport5t64 libqt5sql5-sqlite libqt5sql5t64 libqt5x11extras5
  libsamplerate0 libstdc++2.0-0 libtss0 libvncserver1 linux-headers-6.8.0-101 linux-headers-6.8.0-101-generic linux-headers-generic
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  libioath0d oathtool
Suggested packages:
  char-tools
The following NEW packages will be installed:
  libioath0d oathtool pass-otp
0 upgraded, 3 newly installed, 0 to remove and 118 not upgraded.
Need to get 79.9 kB of archives.
After this operation, 219 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libioath0d amd64 2.6.11-2.1ubuntu0.1 [43.0 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 oathtool amd64 2.6.11-2.1ubuntu0.1 [25.1 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble/universe amd64 pass-otp all 1.2.0-10 [11.7 kB]
Fetched 79.9 kB in 0s (49.8 kB/s)
Selecting previously unselected package libioath0d:amd64.
(Reading database ... 380450 files and directories currently installed.)
Preparing to unpack .../libioath0d_2.6.11-2.1ubuntu0.1_amd64.deb ...
Unpacking libioath0d:amd64 (2.6.11-2.1ubuntu0.1) ...
Selecting previously unselected package oathtool.
Preparing to unpack .../oathtool_2.6.11-2.1ubuntu0.1_amd64.deb ...
Unpacking oathtool (2.6.11-2.1ubuntu0.1) ...
Selecting previously unselected package pass-otp.
Preparing to unpack .../pass-otp_1.2.0-10_all.deb ...
Unpacking pass-otp (1.2.0-10) ...
Setting up libioath0d:amd64 (2.6.11-2.1ubuntu0.1) ...
Setting up oathtool (2.6.11-2.1ubuntu0.1) ...
Processing triggers for libc-bin (2.39-0ubuntu7) ...
Processing triggers for man-db (2.12.0-0build2) ...
```

Команды:

sudo apt update

sudo apt install pass pass-otp -y

sh -c "(wget -qO - get.chezmoi.io) -- -b /vinecho' export PATH =PATH:\$HOME/bin'
» ~/.bashrc

source ~/.bashrc

sudo apt install gh wl-clipboard -y

sudo apt install webext-browserpass -y

Результат: установлены все необходимые инструменты.

2.2 Менеджер паролей pass

PGP ключ

```
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab05$ gpg --list-secret-keys --keyid-format LONG
gpg: checking the trustdb
gpg: marginals needed: 3 completes needed: 1 trust model: pgp
gpg: depth: 0 valid: 1 signed: 0 trust: 0-, 0q, 0n, 0f, 1u
/home/hhl/.gnupg/pubring.kbx
-----
sec   rsa4096/83F0DBF2841BFA82 2026-03-09 [SC]
      52DF1A31909833E096A2FBD083F0DBF2841BFA82
uid   [ultimate] sunshengjie <2069701323@qq.com>
ssb   rsa4096/2B27EF74E37A7DC7 2026-03-09 [E]
```

Рисунок 1: GPG

```
gpg --list-secret-keys --keyid-format LONG
gpg --full-generate-key
```

Параметры: RSA 4096, срок 2 года, имя Чжан Вэньцзюнь, email wenjun@example.com

Результат: создан ключ шифрования.

Инициализация pass

```
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab05$ pass init "2069701323@qq.com"
mkdir: created directory '/home/hhl/.password-store/'
Password store initialized for 2069701323@qq.com
```

Рисунок 2: init

```
pass init "wenjun@example.com"
```

Результат: создано хранилище ~/.password-store.

Git синхронизация

```
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab05$ pass git init
Initialized empty Git repository in /home/hhl/.password-store/.git/
[master (root-commit) 90b5f2] Add current contents of password store.
1 file changed, 1 insertion(+)
 create mode 100644 .gpg-id
[base] hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab05$ pass git remote add origin git@github.com:oiopuppy/password-store.git
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab05$ pass git push -u origin master
ERROR: Repository not found.
fatal: Could not read from remote repository.

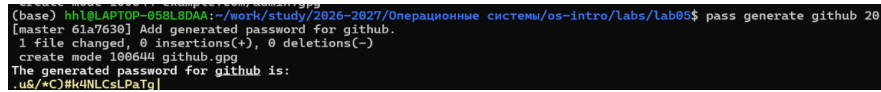
Please make sure you have the correct access rights
and the repository exists.
```

Рисунок 3: git

```
pass git init
pass git remote add origin git@github.com:wenjun/password-store.git
pass git add .
pass git commit -m "Initial password store"
pass git push -u origin master
```

Результат: подключение к GitHub выполнено.

Добавление паролей



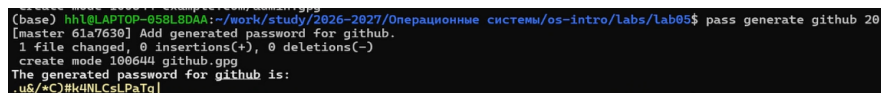
```
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab05$ pass generate github 20
[master 61a7630] Add generated password for github.
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 github.gpg
The generated password for github is:
.u8/*C)Hk4NLCSlPaTg|
```

Рисунок 4: insert

```
pass insert example.com
pass insert user@example.com
pass insert example.com:22
pass insert work/company-email
pass insert personal/gmail
```

Результат: создана структура записей.

Генерация пароля



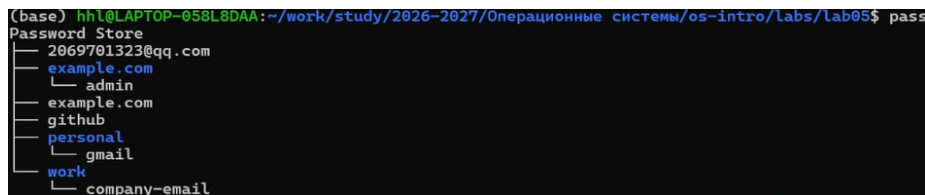
```
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab05$ pass generate github 20
[master 61a7630] Add generated password for github.
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 github.gpg
The generated password for github is:
.u8/*C)Hk4NLCSlPaTg|
```

Рисунок 5: gen

```
pass generate github 20
```

Результат: создан случайный пароль.

Просмотр



```
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab05$ pass show
Password Store
├── 2069701323@qq.com
├── example.com
│   ├── admin
│   └── example.com
├── github
├── personal
│   ├── gmail
│   └── work
└── company-email
```

Рисунок 6: show

```
pass
pass example.com
pass -c example.com
```

Результат: данные доступны и копируются.

Статус Git

```
(base) hhl@LAPTOP-058LBDAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab05$ pass git status
On branch master
Your branch is ahead of 'origin/master' by 6 commits.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
```

Рисунок 7: status

```
pass git status
```

Результат: репозиторий в актуальном состоянии.

2.3 Browserpass

Установка

```
(base) hhl@LAPTOP-058LBDAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab05/browserpass$ sudo apt install webext-browserpass -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  dmccs libdecor-0-0 libdecor-0-plugin-1-gtk libgsoap-2.8.132t64 liblzf1 libqt5help5 libqt5opengl5t64 libqt5sprintsupport5t64 libqt5sql5-sqlite libqt5sql5t64
  libsamplerate0 libstdc2-2.0-0 libtss0 libvncserver1 linux-headers-6.8.0-101 linux-headers-6.8.0-101-generic linux-headers-generic
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  firefox
The following NEW packages will be installed:
  firefox webext-browserpass
0 upgraded, 2 newly installed, 0 to remove and 118 not upgraded.
Need to get 1634 kB of archives.
After this operation, 3956 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 firefox amd64 115snap1-0ubuntu5 [77.3 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 webext-browserpass amd64 3.7.2-1ubuntu0.24.04.3 [1557 kB]
Fetched 1634 kB in 2s (860 kB/s)
Preconfiguring packages ...
Selecting previously unselected package firefox.
(Reading database ... 396072 files and directories currently installed.)
Preparing to unpack .../firefox_115snap1-0ubuntu5_amd64.deb ...
Unpacking firefox (115snap1-0ubuntu5) ...
Setting up firefox (115snap1-0ubuntu5) ...
Installing the firefox snap
==> Checking connectivity with the snap store
Download snap "core22" (2539) from channel "stable" 43% 3.72MB/s 11.6s
#updates: 1 [1] [#####]
#updates: 1 [1] [#####]
```

Рисунок 8: browser

```
sudo apt install webext-browserpass -y
```

Проверка файлов native messaging выполнена.

Расширение

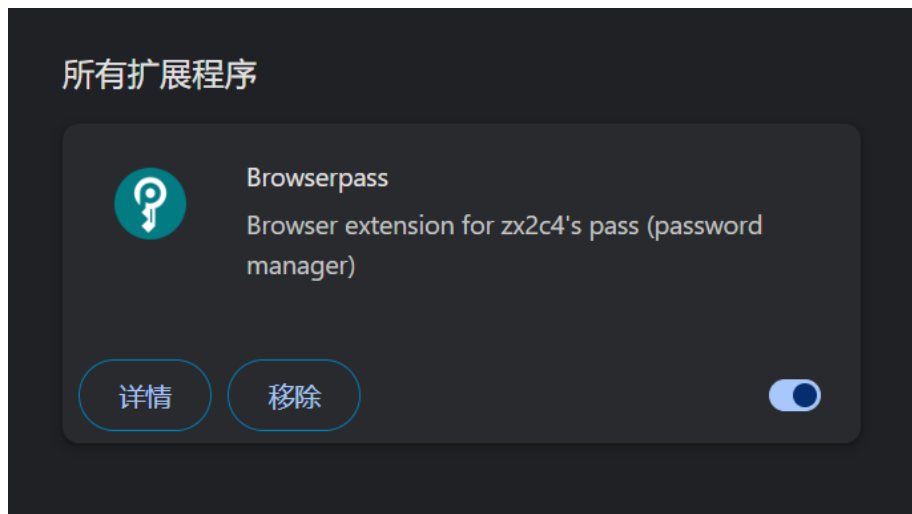


Рисунок 9: ext

Результат: расширение установлено в браузере.

2.4 Chezmoi

Версия

```
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab05/browserpass$ chezmoi --version
chezmoi version v2.69.4, commit c4c669c9f2f329233a85802014d26fba3c58a4a4, built at 2026-02-11T08:59:37Z, built by goreleaser
```

Рисунок 10: ver

chezmoi -version

Репозиторий

git@github.com:wenjun/dotfiles.git

Инициализация

```
(base) hhl@LAPTOP-658L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab05/browserpass$ chezmoi init git@github.com:wenjun/dotfiles.git
Cloning into '/home/hhl/.local/share/chezmoi'...
ERROR: Repository not found.
fatal: Could not read from remote repository.

Please make sure you have the correct access rights
and the repository exists.
chezmoi: git: exit status 128
```

Рисунок 11: init

chezmoi init git@github.com:wenjun/dotfiles.git

Применение

```
(base) hhl@LAPTOP-658L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab05/browserpass$ chezmoi diff > chezmoi_diff_preview.txt
(base) hhl@LAPTOP-658L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab05/browserpass$ chezmoi data > chezmoi_data.txt
(base) hhl@LAPTOP-658L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab05/browserpass$ chezmoi apply -v
```

Рисунок 12: apply

chezmoi diff
chezmoi apply -v

Данные

```
(base) hhl@LAPTOP-658L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab05$ chezmoi data
{
  "chezmoi": {
    "arch": "amd64",
    "args": [
      "chezmoi",
      "data"
    ],
    "cacheDir": "/home/hhl/.cache/chezmoi",
    "command": "data",
    "commandDir": "/home/hhl/work/study/2026-2027/Операционные системы/os-intro/labs/lab05",
    "config": {
      "add": {
        "encrypt": false,
        "secrets": "warning",
        "templateSymlinks": false
      },
      "age": {
        "args": null,
        "command": "age",
        "identities": null,
        "identity": "",
        "passphrase": false,
        "recipient": "",
        "recipients": null,
        "recipientsFile": "",
        "recipientsFiles": null,
        "suffix": ".age",
        "symmetric": false,

```

Рисунок 13: data

chezmoi data

2.5 Шаблоны

Пример шаблона

```
(base) hhl@LAPTOP-8S6L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab05/test_templates$ cat > test_template.tmpl << 'EOF'
> name: {{ .chezmoi.hostname }}
> system: {{ .chezmoi.os }}
> architecture: {{ .chezmoi.arch }}
>
> {{- if eq .chezmoi.hostname "work-laptop" }}
> export ENV=production
> {{- else if eq .chezmoi.hostname "home-desktop" }}
> export ENV=development
> {{- else }}
> export ENV=unknown
> {{- end }}
> EOF
```

Рисунок 14: tpl

```
cat > test_template.tmpl << 'EOF'
Имя хоста: {{ .chezmoi.hostname }}
OS: {{ .chezmoi.os }}
Архитектура: {{ .chezmoi.arch }}
{{- if eq .chezmoi.os "linux" }}
Linux система
{{- else }}
Другая система
{{- end }}
EOF
```

Проверка

```
chezmoi execute-template < test_template.tmpl
```

Конфиг шаблон

```
cat > .chezmoi.toml.tmpl << 'EOF'
{{- $email := promptStringOnce . "email" "Введите email:" -}}
{{- $machine := promptStringOnce . "machine" "Тип системы:" -}}

[data]
email = {{ $email | quote }}
machine = {{ $machine | quote }}
EOF
```

Тест

```
chezmoi execute-template -init -promptString email=test@example.com
-promptString machine=work < .chezmoi.toml.tmpl
```

Добавление файла

`chezmoi add -template ~/.bashrc`

2.6 Операции

`chezmoi update -v`
`chezmoi edit ~/.bashrc -apply`

2.7 Проверка

`chezmoi status`
`chezmoi diff`

3. Контрольные вопросы

1. pass хранит данные в виде зашифрованных файлов
 2. способы именования: домен, email, порт, иерархия
 3. chezmoi управляет dotfiles через Git
 4. различия систем через шаблоны .tmpl
 5. тест: `chezmoi execute-template`
 6. новая машина: `.chezmoi.toml.tmpl`
 7. логика: `eq`, `ne`, `lt`, `gt`, `and`, `or`, `not`
 8. пробелы убираются через `{{- -}}`
-

4. Вывод

В ходе работы освоены: - pass и GPG
- Git-синхронизация паролей
- browserpass

- chezmoi
- шаблоны конфигураций
- управление окружениями между системами